

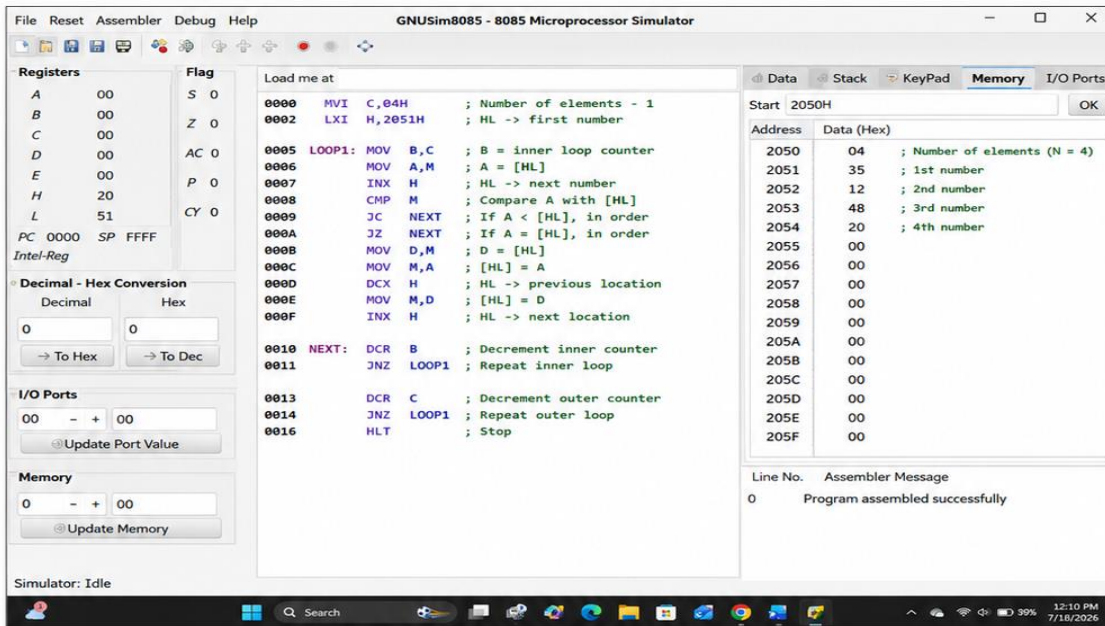
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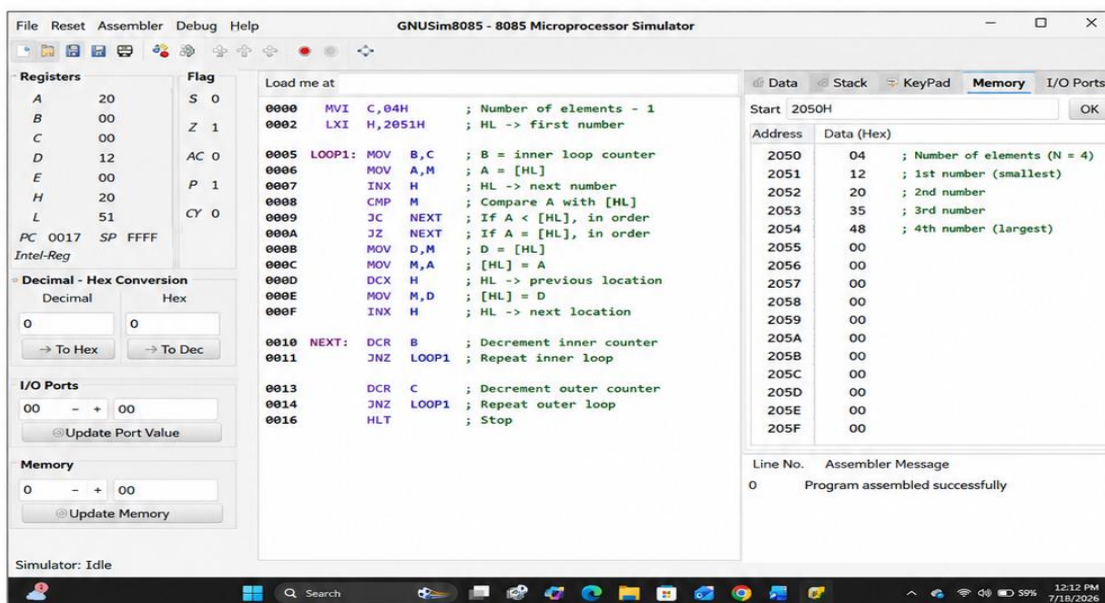
8085 processor : 16 to 23

16. Write an assembly language program to arrange numbers in Ascending order.

Program 16: Arrange Numbers in Ascending Order (8085)



Before Execution (Unsorted Numbers)



After Execution (Sorted Numbers in Ascending Order)

Observation:

Before execution, the numbers were 35H, 12H, 48H, 20H.

After execution, the numbers are arranged in ascending order as 12H, 20H, 35H, 48H.

17. Write an assembly language program to arrange numbers in Descending order.

17. Arrange Numbers in Descending Order (8085)

Program : To arrange numbers in Descending order.

Before Execution (Unsorted Numbers)

Registers: A: 00, B: 00, C: 00, D: 00, E: 00, H: 20, L: 51, PC: 0000, SP: FFFF, Int-Reg: S: 0, Z: 0, AC: 0, P: 0, CY: 0.

Memory (Address 2050H):

Address	Data (Hex)	Comment
2050	04	Number of elements (N = 4)
2051	35	1st number
2052	12	2nd number
2053	48	3rd number
2054	20	4th number
2055	00	
2056	00	
2057	00	
2058	00	

After Execution (Sorted Numbers in Descending Order)

Registers: A: 48, B: 00, C: 00, D: 20, E: 00, H: 20, L: 55, PC: 0017, SP: FFFF, Int-Reg: S: 0, Z: 0, AC: 0, P: 0, CY: 1.

Memory (Address 2050H):

Address	Data (Hex)	Comment
2050	04	Number of elements (N = 4)
2051	48	1st number (largest)
2052	35	2nd number
2053	20	3rd number
2054	12	4th number (smallest)
2055	00	
2056	00	
2057	00	
2058	00	

Observation:

Before execution, the numbers were : 35H, 12H, 48H, 20H.

After execution, the numbers are arranged in Descending order as : 48H, 35H, 20H, 12H.

18. Write an assembly language program to find the largest number in an array.

18. Write an assembly language program to find the largest number in an array (8085).

Before Execution

Registers: A=00, B=00, C=00, D=00, E=00, H=00, L=00, PC=0000, SP=FFFF, Int-Reg=00. Flags: S=0, Z=0, AC=0, P=0, CY=0.

```

2000 MVI C,05H ; Number of elements (N = 5)
2002 LXI H,2051H ; HL -> first element
2005 MOV A,M ; A = first element (assume largest)
2006 INX H ; HL -> next element
2007 DCR C ; Decrement count (N-1)

2008 LOOP: CMP ; Compare A with [HL]
2009 JNC NEXT ; If A >= [HL], no change
200B MOV A,M ; Else, A = [HL] (new largest)
200C NEXT: INX H ; HL -> next element
200D DCR C ; Decrement count
200E JNZ LOOP ; Repeat until all compared
2010 STA 2060H ; Store largest number at 2060H
2013 HLT ; Stop
  
```

Memory (Start 2050H):

Address	Data (Hex)	Comment
2050	05	; Number of elements (N = 5)
2051	13	; 1st element
2052	2F	; 2nd element
2053	07	; 3rd element
2054	8A	; 4th element
2055	3E	; 5th element
2056	00	
2057	00	
2058	00	
2059	00	
205A	00	
205B	00	
205C	00	
205D	00	
205E	00	
205F	00	
2060	--	; Location to store largest number

Assembler Message: Program assembled successfully.

After Execution

Registers: A=8A, B=00, C=00, D=00, E=00, H=20, L=56, PC=2014, SP=FFFF, Int-Reg=00. Flags: S=0, Z=0, AC=0, P=0, CY=0.

```

2000 MVI C,05H ; Number of elements (N = 5)
2002 LXI H,2051H ; HL -> first element
2005 MOV A,M ; A = first element (assume largest)
2006 INX H ; HL -> next element
2007 DCR C ; Decrement count (N-1)

2008 LOOP: CMP ; Compare A with [HL]
2009 JNC NEXT ; If A >= [HL], no change
200B MOV A,M ; Else, A = [HL] (new largest)
200C NEXT: INX H ; HL -> next element
200D DCR C ; Decrement count
200E JNZ LOOP ; Repeat until all compared
2010 STA 2060H ; Store largest number at 2060H
2013 HLT ; Stop
  
```

Memory (Start 2050H):

Address	Data (Hex)	Comment
2050	05	; Number of elements (N = 5)
2051	13	; 1st element
2052	2F	; 2nd element
2053	07	; 3rd element
2054	8A	; 4th element
2055	3E	; 5th element
2056	00	
2057	00	
2058	00	
2059	00	
205A	00	
205B	00	
205C	00	
205D	00	
205E	00	
205F	00	
2060	8A	; Largest number stored here

Assembler Message: Program assembled successfully.

Input (Before Execution) :
 Number of elements (N) = 5
 Array elements (starting from 2051H) = 13H, 2FH, 07H, 8AH, 3EH
Output (After Execution) :
 Largest number = 8AH (stored at memory location 2060H)

19. Write an assembly language program to find the Minimum number in an array.

19. Write an Assembly Language Program to Find the Minimum Number in an Array (8085).

Before Execution (Unsorted Array)

Registers: A=00, B=00, C=00, D=00, E=00, H=20, L=51, PC=2000, SP=FFFF, Int-Reg=00. Flags: S=0, Z=0, AC=0, P=0, CY=0.

Program Code:

```

2000 MVI C,05H ; Number of elements (N = 5)
2002 LXI H,2051H ; HL -> first element
2005 MOV A,M ; A <- first element (assume minimum)
2006 INX H ; HL -> next element
2007 DCR C ; Decrement count (N-1)

2008 LOOP: CMP ; Compare A with [HL]
2009 JNC NEXT ; If A <= [HL], no change
200B MOV A,M ; Else, A <- [HL] (new minimum)
200C NEXT: INX H ; HL -> next element
200D DCR C ; Decrement count
200E JNZ LOOP ; Repeat until all compared
2010 STA 2060H ; Store minimum at 2060H
2013 HLT ; Stop
  
```

Memory (2050H):

Address	Data (Hex)	Comment
2050	05	; Number of elements (N = 5)
2051	13	; 1st element
2052	2F	; 2nd element
2053	07	; 3rd element
2054	8A	; 4th element
2055	3E	; 5th element
2056	00	
2057	00	
2058	00	
2059	00	
205A	00	
205B	00	
205C	00	
205D	00	
205E	00	
205F	00	
2060	--	; Location to store minimum

Assembler Message: Program assembled successfully

After Execution (Minimum Number Stored)

Registers: A=07, B=00, C=00, D=00, E=00, H=20, L=56, PC=2014, SP=FFFF, Int-Reg=00. Flags: S=0, Z=0, AC=0, P=0, CY=0.

Program Code (same as before):

```

2000 MVI C,05H ; Number of elements (N = 5)
2002 LXI H,2051H ; HL -> first element
2005 MOV A,M ; A <- first element (assume minimum)
2006 INX H ; HL -> next element
2007 DCR C ; Decrement count (N-1)

2008 LOOP: CMP ; Compare A with [HL]
2009 JNC NEXT ; If A <= [HL], no change
200B MOV A,M ; Else, A <- [HL] (new minimum)
200C NEXT: INX H ; HL -> next element
200D DCR C ; Decrement count
200E JNZ LOOP ; Repeat until all compared
2010 STA 2060H ; Store minimum at 2060H
2013 HLT ; Stop
  
```

Memory (2050H):

Address	Data (Hex)	Comment
2050	05	; Number of elements (N = 5)
2051	13	; 1st element
2052	2F	; 2nd element
2053	07	; 3rd element
2054	8A	; 4th element
2055	3E	; 5th element
2056	00	
2057	00	
2058	00	
2059	00	
205A	00	
205B	00	
205C	00	
205D	00	
205E	00	
205F	00	
2060	07	; Minimum number stored here

Assembler Message: Program assembled successfully

Input (Before Execution) :
 Number of elements (N) = 5
 Array elements (starting from 2051H) = 13H, 2FH, 07H, 8AH, 3EH

Output (After Execution) :
 Minimum number = 07H (stored at memory location 2060H)

20. Write an Assembly Language Program to Find the LCM of Two Numbers (8085).

Before Execution (Input)

File **Reset** **Assembler** **Debug** **Help**

Registers **Flag**

A	00	S	0
B	00	Z	1
C	00		
D	00	AC	0
E	01		
H	00	P	1
L	06	CY	0

PC 2030 SP FFFF
Int-Reg 00

Decimal - Hex Conversion

Decimal	Hex
0	0
→ To Hex	→ To Dec

I/O Ports

00	- + 00
⊗ Update Port Value	

Memory

00	- + 00
⊗ Update Memory	

Simulator: Idle

After Execution (LCM Stored)

GNUSim8085 - 8085 Microprocessor Simulator

Load me at

2000	MVI	C, 02H	; First number = 02H (2)	
2002	MVI	D, 03H	; Second number = 03H (3)	
2004	MOV	A, C	; A = first number	
2005	MOV	B, D	; B = second number	
2006	DAD	B	; HL = A + B	
2007	MOV	H, D	; H = second number	
2008	LDC	L, C	; L = first number	
2009			; Compare H and L	
200A	JZ	STORE	; If equal, GCD found	
200B	3C	SMALL	; If H < L, go to SMALL	
200C	SUB	L	; H = H - L	
200D	JMP	LOOP		
200E	SMALL:	CMP	H	; Compare L and H
200F	JZ	SCR		
2010	3C	BIG		; If L < H, go to BIG
2011	SUB	H		; L = L - H
2012	JMP	SMALL		
2013		SUB	H	
2014	JMP	BIG		
2015	STORE:	MOV	E, H	; GCD in H
2016	MVI	D, 00H		
2018	MOV	A, C		
2019	MOV	L, E		; L = GCD
201A	MOV	H, D		; HL = (First x Second)
201B	DAD	B		; Divide by GCD
201C	MOV	B, L		
201D	MOV	C, H		
201E	MOV	A, B		
201F	DCR	A		
2020	MOV	A, B		; Loop for division
2021	DIV	MOV	A, C	
2022	SBB			
2023	MOV	C, A		; Subtract GCD

Data	Stack	KeyPad	Memory	I/O Ports
Start 2050H				OK
Address	Data (Hex)	Comment		
2050	06	; LCM of 02H and 03H = 06H		
2051	02	; First number = 02H (2)		
2052	03	; Second number = 03H (3)		
2053	00			
2054	00			
2055	00			
2056	00			
2057	00			
2058	00			
205A	00			
205B	00			
205C	00			
205D	00			
205E	00			
205F	00			
2060	--			

Line No.	Assembler Message
0	Program assembled successfully

Output (After Execution) : LCM = 06H (6)

LCM is stored at memory location 2050H.

21. Write an assembly language program to find the GCD of 2 numbers.

21. Write an Assembly Language Program to Find the GCD of Two Numbers (8085).

Before Execution (Input)

File Reset Assembler Debug Help

GNUSim8085 - 8085 Microprocessor Simulator

Registers

A	00
B	00
C	00
D	00
E	00
H	20
L	50
PC	2000
SP	FFFF
Int-Reg	00

Flag

S	0
Z	0
AC	0
P	0
CY	0

Decimal - Hex Conversion

Decimal	Hex
0	0

I/O Ports

00	-	00
----	---	----

Memory

0	-	00
---	---	----

Load me at

```

2000 MVI B,15H ; First number = 15H (21)
2002 MVI C,09H ; Second number = 09H (9)
2004 MOV A,B ; A = First number
2005 MOV D,C ; D = Second number
2006 LOOP: CMP D ; Compare A and D
2007 JZ DONE ; If equal, GCD found
2009 JC L1 ; If A < D, go to L1
200A SUB D ; A = A - D
200B JMP LOOP ; Repeat loop
200D L1: MOV A,D ; D = D - B
200E SUB B ; D = D - B
200F MOV D,A ; Update D
2010 JMP LOOP
2013 DONE: MOV E,A ; GCD in A -> store in E
2014 STA 2050H ; Store GCD at 2050H
2017 HLT ; Stop

```

Start 2050H

Address	Data (Hex)	Comment
2050	00	; Location to store GCD
2051	15	; First number = 15H (21)
2052	09	; Second number = 09H (9)
2053	00	
2054	00	
2055	00	
2056	00	
2057	00	
2058	00	
2059	00	
205A	00	
205B	00	
205C	00	
205D	00	
205E	00	
205F	00	
2060	--	

Line No. Assembler Message

0 Program assembled successfully

Simulator: Idle

After Execution (GCD Stored)

File Reset Assembler Debug Help

GNUSim8085 - 8085 Microprocessor Simulator

Registers

A	03
B	15
C	09
D	03
E	03
H	20
L	50
PC	2018
SP	FFFF
Int-Reg	00

Flag

S	0
Z	1
AC	0
P	1
CY	0

Decimal - Hex Conversion

Decimal	Hex
0	0

I/O Ports

00	-	00
----	---	----

Memory

0	-	00
---	---	----

Load me at

```

2000 MVI B,15H ; First number = 15H (21)
2002 MVI C,09H ; Second number = 09H (9)
2004 MOV A,B ; A = First number
2005 MOV D,C ; D = Second number
2006 LOOP: CMP D ; Compare A and D
2007 JZ DONE ; If equal, GCD found
2009 JC L1 ; If A < D, go to L1
200A SUB D ; A = A - D
200B JMP LOOP ; Repeat loop
200D L1: MOV A,D ; D = D - B
200E SUB B ; D = D - B
200F MOV D,A ; Update D
2010 JMP LOOP
2013 DONE: MOV E,A ; GCD in A -> store in E
2014 STA 2050H ; Store GCD at 2050H
2017 HLT ; Stop

```

Start 2050H

Address	Data (Hex)	Comment
2050	03	; GCD = 03H (3)
2051	15	; First number = 15H (21)
2052	09	; Second number = 09H (9)
2053	00	
2054	00	
2055	00	
2056	00	
2057	00	
2058	00	
2059	00	
205A	00	
205B	00	
205C	00	
205D	00	
205E	00	
205F	00	
2060	--	

Line No. Assembler Message

0 Program assembled successfully

Simulator: Idle

Input (Before Execution) : First Number = 15H (21), Second Number = 09H (9)

Output (After Execution) : GCD = 03H (3)

GCD is stored at memory location 2050H.

22. Write an assembly language program to find factorial of n in the given number.

22. Write an Assembly Language Program to Find Factorial of N in the Given Number (8085).

Before Execution (Input)

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The registers section shows A=00, B=00, C=00, D=00, E=00, H=20, L=50, PC=2000, SP=FFFF, and Int-Reg=00. The flags section shows S=0, Z=0, AC=0, P=0, and CY=0. The memory section shows the assembly code for calculating the factorial of 5, starting at address 2050H. The code includes instructions for loading N=05, initializing result=01, and a loop to calculate the factorial.

Address	Data (Hex)	Comment
2050	05	; N (Input) = 05 (5)
2051	00	; Low byte of factorial
2052	00	; High byte of factorial
2053	00	
2054	00	
2055	00	
2056	00	
2057	00	
2058	00	
2059	00	
205A	00	
205B	00	
205C	00	
205D	00	
205E	00	
205F	--	

After Execution (Factorial Stored)

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface after execution. The registers section shows A=00, B=78, C=00, D=00, E=00, H=00, L=78, PC=2027, SP=FFFF, and Int-Reg=00. The flags section shows S=0, Z=1, AC=0, P=1, and CY=0. The memory section shows the assembly code for calculating the factorial of 5, starting at address 2050H. The code includes instructions for loading N=05, initializing result=01, and a loop to calculate the factorial.

Address	Data (Hex)	Comment
2050	05	; N (Input) = 05 (5)
2051	78	; Low byte of factorial
2052	00	; High byte of factorial
2053	00	
2054	00	
2055	00	
2056	00	
2057	00	
2058	00	
2059	00	
205A	00	
205B	00	
205C	00	
205D	00	
205E	00	
205F	--	

Input (Before Execution) : N = 05 (5)
Output (After Execution) : Factorial of 5 = 120 = 0078H
Result stored at memory location 2051H (Low byte) and 2052H (High byte).

23. Write a program to convert Decimal number to Hexadecimal number.

23. Write a program to convert Decimal number to Hexadecimal number (8085).

Before Execution (Input)

After Execution (Hexadecimal Result Stored)

Input (Decimal number) : 77 (4DH)
Output (Hexadecimal) : 4D (High nibble = 04H, Low nibble = 0DH)
Result stored at 2050H = 04H and 2051H = 0DH.